

Program-10 : Demonstrate Kubernetes Dashboard and CLI for Cluster Monitoring

Description :

This program demonstrates how to monitor and manage a Kubernetes cluster using both the Kubernetes Dashboard (Web UI) and kubectl CLI. The Dashboard provides a visual interface, while kubectl allows command-line monitoring and control of cluster resources.

Step-1 : Start Minikube with CPU and Memory

```
minikube start --driver=docker --cpus=2 --memory=4096
```

Step-2 : Enable Kubernetes Dashboard

```
minikube dashboard
```

Step-3 : Create Project Directory

```
cd ~  
mkdir program-10  
cd program-10
```

Step-4 : Create Deployment YAML File

```
nano deployment.yaml
```

```
apiVersion: apps/v1  
kind: Deployment  
metadata:  
  name: nginx-deployment  
  labels:  
    app: nginx  
spec:  
  replicas: 2  
  selector:  
    matchLabels:  
      app: nginx  
  template:  
    metadata:  
      labels:  
        app: nginx  
    spec:  
      containers:  
        - name: nginx  
          image: nginx:latest  
          ports:  
            - containerPort: 80
```

Step-5 : Create Service YAML File

```
nano service.yaml
```

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
```

Step-6 : Apply Deployment

```
kubectl apply -f deployment.yaml
```

Step-7 : Apply Service

```
kubectl apply -f service.yaml
```

Step-8 : Monitor via CLI

```
kubectl get pods
```

```
kubectl get deployments
```

```
kubectl get svc
```

Step-9 : Check Nodes

```
kubectl get nodes
```

Step-10 : Check Pod Logs

```
kubectl logs <pod-name>
```

Step-11 : Describe Pod

```
kubectl describe pod <pod-name>
```

```

pavan-k@pavank1RV24MC072:~$ mkdir program-10
cd program-10
pavan-k@pavank1RV24MC072:~/program-10$ nano deployment.yaml
pavan-k@pavank1RV24MC072:~/program-10$ nano service.yaml
pavan-k@pavank1RV24MC072:~/program-10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
pavan-k@pavank1RV24MC072:~/program-10$ kubectl apply -f service.yaml
service/nginx-service created
pavan-k@pavank1RV24MC072:~/program-10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-wlm2n   1/1     Running   0           15s
nginx-deployment-6f9664446b-xxn74   1/1     Running   0           15s
program-8                            1/1     Running   2 (2m53s ago)  41h
pavan-k@pavank1RV24MC072:~/program-10$ kubectl get deployments
NAME            READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment 2/2     2            2           22s
pavan-k@pavank1RV24MC072:~/program-10$ kubectl get svc
NAME            TYPE        CLUSTER-IP      EXTERNAL-IP  PORT(S)          AGE
kubernetes      ClusterIP   10.96.0.1        <none>       443/TCP          41h
nginx-service    NodePort    10.108.193.31    <none>       80:30932/TCP     60s
pavan-k@pavank1RV24MC072:~/program-10$ kubectl get nodes
NAME        STATUS    ROLES    AGE   VERSION
minikube    Ready    control-plane  41h   v1.34.0
pavan-k@pavank1RV24MC072:~/program-10$ kubectl logs nginx-deployment-6f9664446b-wlm2n
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/01/06 14:15:16 [notice] 1#1: using the "epoll" event method
2026/01/06 14:15:16 [notice] 1#1: nginx/1.29.4
2026/01/06 14:15:16 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/01/06 14:15:16 [notice] 1#1: OS: Linux 6.14.0-27-generic
2026/01/06 14:15:16 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/01/06 14:15:16 [notice] 1#1: start worker processes
2026/01/06 14:15:16 [notice] 1#1: start worker process 29
2026/01/06 14:15:16 [notice] 1#1: start worker process 30
2026/01/06 14:15:16 [notice] 1#1: start worker process 31
2026/01/06 14:15:16 [notice] 1#1: start worker process 32
2026/01/06 14:15:16 [notice] 1#1: start worker process 33
2026/01/06 14:15:16 [notice] 1#1: start worker process 34
2026/01/06 14:15:16 [notice] 1#1: start worker process 35
2026/01/06 14:15:16 [notice] 1#1: start worker process 36
2026/01/06 14:15:16 [notice] 1#1: start worker process 37
2026/01/06 14:15:16 [notice] 1#1: start worker process 38

```

Step-12 : Access Application via Minikube

minikube service nginx-service

Welcome to nginx!

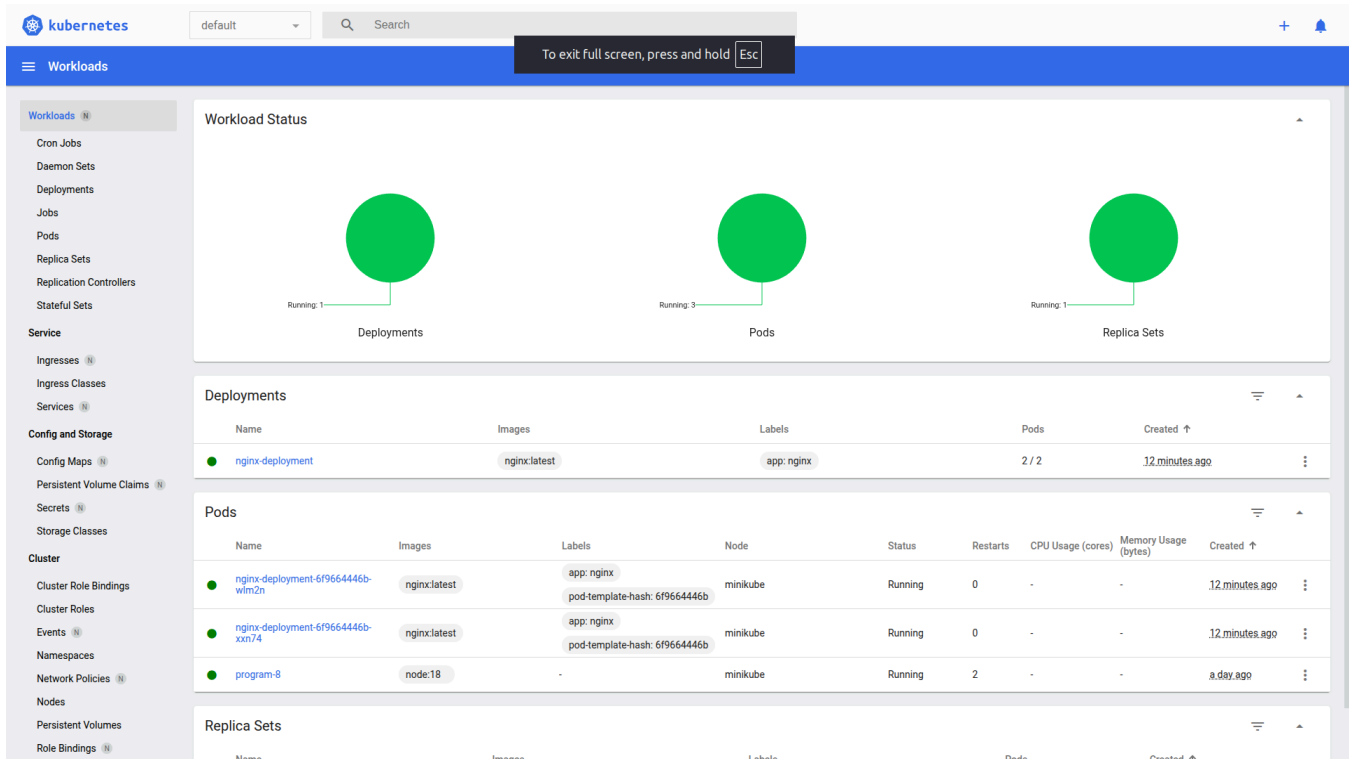
If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Step-13 : Open Kubernetes Dashboard

minikube dashboard



Step-14 : Cleanup Process

kubectl delete svc --all

kubectl delete deployments --all

Result :

The Kubernetes cluster was successfully monitored using both the Kubernetes Dashboard and kubectl CLI. Deployments, pods, services, logs, and node status were inspected using command-line and web-based monitoring tools.